Supplementary material

Highly efficient lithium extraction from magnesium-rich brines with ionic liquid-based collaborative extractants: Thermodynamics and molecular insights

Gangqiang Yu[a,b], Xinhe Zhang[a], Tobias Hubach[b], Biaohua Chen[a,*], Christoph Held[b,*]

[a]*Faculty of Environment and Life, Beijing University of Technology, 100 Ping Le Yuan, Chaoyang District, Beijing, 100124, China*

[b]*Laboratory of Thermodynamics, Department of Biochemical and Chemical Engineering, TU Dortmund University, Emil–Figge–Str. 70, 44227 Dortmund, Germany*

***Corresponding authors**: chenbh@bjut.edu.cn (B. Chen), christoph.held@tu–dortmund.de (C. Held)

The numbers of pages, figures and tables are 20, 5 and 10, respectively.

Table Captions

Table S1. Effects of different ligands on extraction and separation performance of Li^+ in the presence of 0.1 mol/L [BMIM][Tf$_2$N] (Mg^{2+}/Li^+ =40:1; O/A = 3:1; pH = 7; and C_{IL} = 0.1 mol/L) at 298.15 K.

Table S2. Effects of different IL on extraction and extraction performance of Li^+ with TOP as neutral ligand (Mg^{2+}/Li^+ =40:1; O/A = 3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

Table S3. Effects of pH values on extraction efficiency and separation performances (D_{Li+}, and $\beta_{Li+/Mg2+}$) of Li^+/Mg^{2+} with [HOEMIM][Tf$_2$N] + TOP as extractant system (Mg^{2+}/Li^+ =40:1; O/A = 3:1 and C_{IL} = 0.1 mol/L) at 298.15 K.

Table S4. Effects of IL concentrations (C_{IL}) in the organic phase on extraction efficiency and separation performances (D_{Li+}, and $\beta_{Li+/Mg2+}$) of Li/Mg with [HOEMIM][Tf$_2$N] + TOP as extractant system (Mg^{2+}/Li^+ =40:1; O/A = 3:1 and pH = 7) at 298.15 K.

Table S5. Effects of the volume ratio for the organic to aqueous phase (O/A) on extraction efficiency and separation performances (D_{Li+}, and $\beta_{Li+/Mg2+}$) of Li/Mg with [HOEMIM][Tf$_2$N] + TOP as extractant system (Mg^{2+}/Li^+ =40:1; and pH = 7 and C_{IL} = 0.09 mol/L) at 298.15 K.

Table S6. Details experimental data and results predicted by the ePC-SAFT model for effects of the volume ratio for the organic to aqueous phase (O/A) on Li^+ and Mg^{2+} extraction efficiency with [HOEMIM][Tf$_2$N] + TOP as extraction system (Mg^{2+}/Li^+ = 40:1; and pH = 7 and C_{IL} = 0.09 mol/L).

Table S7. Extraction efficiency and concentration of Li^+ (C_{Li+}) in aqueous phase at each stage with [HOEMIM][Tf$_2$N] + TOP an extractant system (Mg^{2+}/Li^+ =40:1; O/A=3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

Table S8. Effects of LiCl concentration on metal washing efficiency (O/A=3:1; 0.5mol/L NaCl) at 298.15 K.

Table S9. Effects of different O/A volume ratio on stripping efficiency of Li^+ at 298.15 K.

Table S10. Extraction performances of with the regenerated organic extractant system at 298.15 K.

Figure Captions

Fig. S1. Physicochemical properties of [HOEMIM][Tf_2N] for (a) TGA, (b) viscosity and (c) density.

Fig. S2. Extraction rate (A) and concentration of Li^+ (C_{Li+}) in aqueous phase (B) at each stage with [HOEMIM][Tf_2N] + TOP an extractant system (Mg^{2+}/Li^+ =40:1; O/A=3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

Fig. S3. UV Spectra (A) and $[HOEMIM]^+$ concentrations (B) of the aqueous phase with and without metal ions after extraction with [HOEMIM][Tf_2N] + TOP as extractant system at 298.15 K.

Fig. S4. FTIR spectra of the organic phase before and after extraction at 298.15 K.

Fig. S5. Plot of $\log D_{Li+} + \log[HOEMIM]^+_{aq} - \log[HOEMIM]^+_{org}$ as a function of log[TOP] (the Li^+ concentration of 0.1 mol/L, O/A = 3:1, C_{IL} = 0.1 mol/L and pH=7) at 298.15 K.

Section I: Washing experiment

All washing experiments were carried out in a 50 mL centrifuge The extracted organic phase was taken 15mL, into which adding a certain concentration of NaCl and LiCl solution of 5mL. The mixture was shaken and subjected to centrifugal extraction at 300 r/min for 30 minutes to achieve washing equilibrium. Subsequently, it was centrifuged at 8000 r/min for 5 minutes to attain complete separation of the aqueous and organic phases. The concentrations of metal ions prior to and following the washing process were measured using an inductively coupled plasma emission spectrometer (ICP-OES Optima 8000). The washing efficacy (E_W) of metal ions were determined by utilizing the following formula:

$$E_{\mathrm{w}} = \frac{C_{\mathrm{w},0} - C_{\mathrm{w},1}}{C_{\mathrm{w},0}} \times 100\% \tag{S1}$$

where $C_{\mathrm{w},0}$ and $C_{\mathrm{w},1}$ represents the concentration of metal ions in the aqueous phase before and after washing, respectively, by measuring experimentally.

Section II: Stripping experiments

All stripping experiments were carried out in a 50 mL centrifuge tube. The washed organic phase was taken 5mL, into which adding a certain amount of 0.5 mol/L HCl. The mixture was shaken and subjected to centrifugal extraction at 300 r/min for 30 minutes to achieve stripping equilibrium. Subsequently, it was centrifuged at 8000 r/min for 5 minutes to attain complete separation of the aqueous and organic phases. The concentrations of metal ions prior to and following the stripping process were measured using an inductively coupled plasma emission spectrometer (ICP-OES Optima 8000). The washing efficacy (E_S) of metal ions were determined by utilizing the following formula:

$$E_{\mathrm{s}} = \frac{C_{\mathrm{s},0} - C_{s,1}}{C_{s,0}} \times 100\% \tag{S2}$$

where $C_{\mathrm{s},0}$ and $C_{\mathrm{s},1}$ represents the concentration of metal ions in the aqueous phase before and after stripping, respectively, by measuring experimentally.

Table S1. Effects of different ligands on extraction and separation performance of Li^{+} in the presence of 0.1 mol/L [BMIM][Tf_2N] (Mg^{2+}/Li^{+} =40:1; O/A = 3:1; pH = 7; and C_{IL} = 0.1 mol/L) at 298.15 K.

Ligands	E_{Li} **(%)**	E_{Mg} **(%)**	D_{Li}	D_{Mg}	$\beta_{Li/Mg}$	**Before extraction**						**After extraction**			
						Aq. Feed				**Extractant**		**Aq. Phase**			
						w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)	w_TOP (%)	w_IL (%)	w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)
C923	4.41	23.84	0.0153	0.1043	0.15	0.34	32.75	0.055	8.36	95.32	4.68	0.35	27.07	0.057	6.91
DBP	8.44	16.12	0.0307	0.0640	0.48	0.34	32.75	0.055	8.36	96.22	3.78	0.32	29.03	0.053	7.41
P204	9.21	29.96	0.0338	0.1426	0.24	0.34	32.79	0.055	8.37	95.86	4.14	0.34	25.46	0.055	6.5
TBP	72.27	2.77	0.8688	0.0095	91.58	0.34	32.75	0.055	8.36	95.78	4.22	0.09	32.24	0.015	8.23
TOP	69.81	1.15	0.7708	0.0039	199.63	0.34	32.75	0.055	8.36	95.42	4.58	0.10	32.59	0.017	8.32

Table S2. Effects of different IL on extraction and extraction performance of Li^+ with TOP as neutral ligand (Mg^{2+}/Li^+ =40:1; O/A = 3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

IL	E_{Li} (%)	E_{Mg} (%)	D_{Li}	D_{Mg}	$\beta_{Li/Mg}$	Before extraction						After extraction			
						Aq. Feed				Extractant		Aq. Phase			
						w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)	w_TOP (%)	w_IL (%)	w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)
[OMIM][Tf_2N]	39.51	1.24	0.2177	0.0042	52.00	0.37	32.75	0.061	8.36	94.8	5.2	0.23	32.55	0.037	8.31
[BMIM][Tf_2N]	69.60	1.15	0.7633	0.0039	197.69	0.37	32.75	0.061	8.36	95.42	4.58	0.12	32.59	0.019	8.32
[EMIM][Tf_2N]	73.69	1.62	0.9337	0.0055	206.88	0.37	32.75	0.061	8.36	95.73	4.27	0.10	32.47	0.016	8.29
[BMP][Tf_2N]	72.22	2.67	0.8665	0.0092	179.03	0.37	32.75	0.061	8.36	95.45	4.55	0.10	32.24	0.017	8.23
[HOEMIM][Tf_2N]	80.27	0.57	1.3558	0.0019	709.92	0.37	32.75	0.061	8.36	95.56	4.44	0.07	32.75	0.012	8.36

Table S3. Effects of pH values on extraction efficiency and separation performances (D_{Li+}, and $\beta_{Li+/Mg2+}$) of Li^+/Mg^{2+} with [HOEMIM][Tf_2N] + TOP as extractant system (Mg^{2+}/Li^+ =40:1; O/A = 3:1 and C_{IL} = 0.1 mol/L) at 298.15 K.

pH	E_{Li} (%)	E_{Mg} (%)	D_{Li}	D_{Mg}	$\beta_{Li/Mg}$
1	75.82	1.22	1.05	0.00	253.65
2	80.18	1.43	1.35	0.00	278.11
3	82.63	1.58	1.59	0.01	296.48
4	83.70	1.55	1.71	0.01	327.01
5	83.06	1.30	1.63	0.00	372.43
6	83.43	1.11	1.68	0.00	448.96
7	83.44	0.71	1.68	0.00	709.01

Table S4. Effects of IL concentrations (C_{IL}) in the organic phase on extraction efficiency and separation performances (D_{Li^+}, and $\beta_{Li^+/Mg^{2+}}$) of Li/Mg with [HOEMIM][Tf_2N] + TOP as extractant system (Mg^{2+}/Li^+ =40:1; O/A = 3:1 and pH = 7) at 298.15 K.

[HOEMIM][Tf_2N] (mol/L)	***E_{Li}* (%)**	***E_{Mg}* (%)**	***D_{Li}***	***D_{Mg}***	***$\beta_{Li/Mg}$***	**Before extraction**						**After extraction**			
						Aq. Feed				**Extractant**		**Aq. Phase**			
						w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)	w_TOP (%)	w_IL (%)	w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)
0.00	0.10	0.86	0.0003	0.0027	0.11	0.40	30.95	0.065	7.9	100	0	0.40	30.75	0.065	7.85
0.01	0.00	0.88	0.0003	0.0030	0.10	0.37	30.95	0.061	7.9	99.55	0.45	0.37	30.75	0.061	7.85
0.02	2.52	0.95	0.0086	0.0032	2.69	0.37	30.95	0.061	7.9	99.10	0.9	0.36	30.75	0.059	7.85
0.03	5.57	0.87	0.0197	0.0029	6.70	0.37	30.95	0.061	7.9	98.65	1.35	0.35	30.75	0.057	7.85
0.04	9.85	0.93	0.0364	0.0031	11.59	0.37	30.95	0.061	7.9	98.20	1.8	0.34	30.75	0.055	7.85
0.05	12.99	0.84	0.0498	0.0028	17.56	0.37	30.95	0.061	7.9	97.76	2.24	0.32	30.79	0.053	7.86
0.06	28.15	0.94	0.1306	0.0032	41.11	0.37	30.95	0.061	7.9	97.32	2.68	0.27	30.75	0.044	7.85
0.07	57.26	0.75	0.4465	0.0025	176.83	0.37	30.95	0.061	7.9	96.87	3.13	0.16	30.83	0.026	7.87
0.08	73.47	0.73	0.9230	0.0025	375.70	0.37	30.95	0.061	7.9	96.43	3.57	0.10	30.87	0.016	7.88
0.09	79.90	0.53	1.3248	0.0018	746.58	0.37	30.95	0.061	7.9	96.00	4	0.07	30.91	0.012	7.89
0.10	79.91	0.57	1.3259	0.0019	693.92	0.37	30.95	0.061	7.9	95.56	4.44	0.07	30.91	0.012	7.89
0.11	80.33	0.71	1.3612	0.0024	569.92	0.38	30.95	0.063	7.9	95.13	4.87	0.07	30.87	0.012	7.88
0.12	80.42	0.87	1.3690	0.0029	466.26	0.38	30.95	0.063	7.9	94.69	5.31	0.07	30.83	0.012	7.87

Table S5. Effects of the volume ratio for the organic to aqueous phase (O/A) on extraction efficiency and separation performances (D_{Li+}, and $\beta_{Li+/Mg2+}$) of Li/Mg with [HOEMIM][Tf_2N] + TOP as extractant system (Mg^{2+}/Li^{+} =40:1; and pH = 7 and C_{IL} = 0.09 mol/L) at 298.15 K.

						Before extraction						After extraction			
						Aq. Feed				Extractant		Aq. Phase			
O/A	E_{Li} (%)	E_{Mg} (%)	D_{Li}	D_{Mg}	$\beta_{Li/Mg}$	w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)	w_TOP (%)	w_IL (%)	w_LiCl (%)	w_MgCl2 (%)	w_Li+ (%)	w_Mg2+ (%)
1:1	37.07	0.88	0.5890	0.0089	66.10	0.29	30.95	0.048	7.9	96	4	0.19	30.79	0.031	7.86
2:1	75.07	0.81	1.5060	0.0041	367.69	0.29	30.95	0.048	7.9	96	4	0.07	30.83	0.012	7.87
3:1	83.16	0.66	1.6458	0.0022	742.11	0.29	30.95	0.048	7.9	96	4	0.05	30.87	0.008	7.88
4:1	87.51	1.11	1.7520	0.0028	621.13	0.29	30.95	0.048	7.9	96	4	0.04	30.79	0.006	7.86
5:1	88.63	1.80	1.5597	0.0037	424.75	0.29	30.95	0.048	7.9	96	4	0.04	30.63	0.006	7.82
6:1	88.90	1.97	1.3346	0.0034	397.50	0.29	30.95	0.048	7.9	96	4	0.03	30.59	0.005	7.81

Table S6. Details experimental data and results predicted by the ePC-SAFT model for effects of the volume ratio for the organic to aqueous phase (O/A) on Li^+ and Mg^{2+} extraction efficiency with [HOEMIM][Tf_2N] + TOP as extraction system (Mg^{2+}/Li^+ = 40:1; and pH = 7 and C_{IL} = 0.09 mol/L).

O/A	E_{Li^+} (%)		$E_{Mg^{2+}}$ (%)	
	Experimental data	**Predicted results**	**Experimental data**	**Predicted results**
1:1	37.07	42.54	0.88	0.00
2:1	75.07	76.40	0.81	0.01
3:1	83.16	88.25	0.66	0.03
4:1	87.51	92.55	1.11	0.05
5:1	88.63	94.60	1.80	0.07
6:1	88.90	95.79	1.97	0.09

Table S7. Extraction efficiency and concentration of NaCl (C_{NaCl}) in aqueous phase at each stage with [HOEMIM][Tf_2N] + TOP an extractant system (Mg^{2+}/Li^{+} =40:1; O/A=3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

NaCl concentration (mol/L)	E_{Li} (%)	E_{Mg} (%)
0	12.17	40.77
0.1	18.50	76.42
0.2	27.28	88.64
0.3	37.19	96.16
0.4	44.44	98.72

Table S8. Effects of LiCl concentration on metal washing efficiency (O/A=3:1; 0.5mol/L NaCl) at 298.15 K.

LiCl concentration (mol/L)	E_{Li} (%)	E_{Mg} (%)
0.02	44.03	99.01
0.04	14.27	97.87
0.06	10.35	97.73
0.08	5.29	98.01
0.10	1.77	99.57

Table S9. Effects of different O/A volume ratio on stripping efficiency of Li^+ at 298.15 K.

O/A	$\boldsymbol{E_{Li}}$ **(%)**
4:1	49.19
3:1	84.80
2:1	99.54
1:1	99.90
1:2	99.99

Table S10. Extraction performances of with the regenerated organic extractant system at 298.15 K.

Extraction times	**E_{Li} (%)**
1	83.22
2	82.72
3	81.79
4	81.22
5	80.69
6	79.46

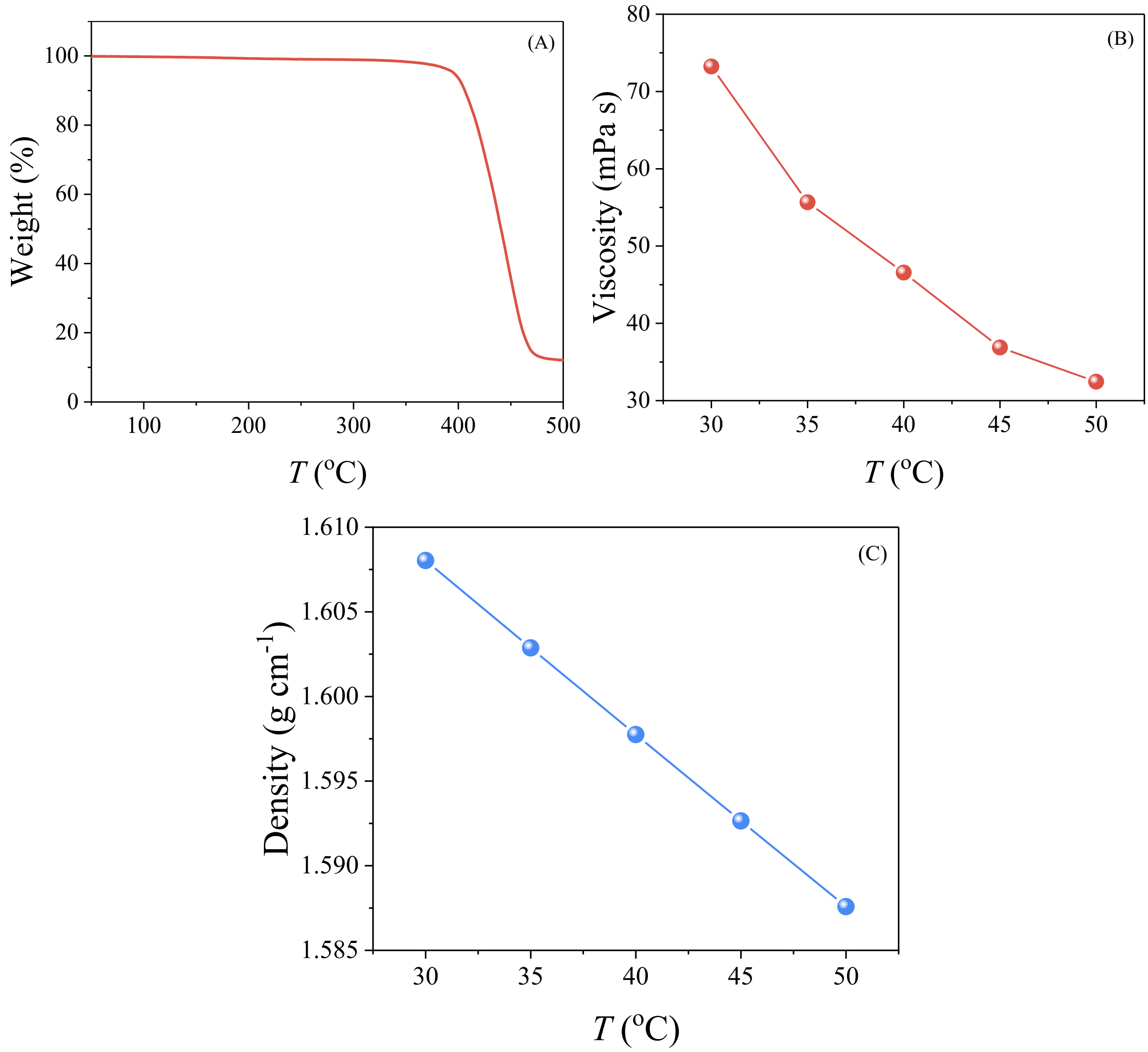

Fig. S1. Physicochemical properties of [HOEMIM][Tf_2N] for (a) TGA, (b) viscosity and (c) density. **Note:** Densities and viscosities of [HOEMIM][Tf_2N] were measured by Anton Paar DMA 4500 M automatic analyzer. The densities with an uncertainty of 0.00005 g/cm^3, and the viscosities with an uncertainty of 1%. Their thermal stabilities were measured at the wide temperature range from 308 to 823 K on a TGA 209 Libra (NETZSCH Germany) at a heating rate of 10 K/min under N_2 atmosphere.

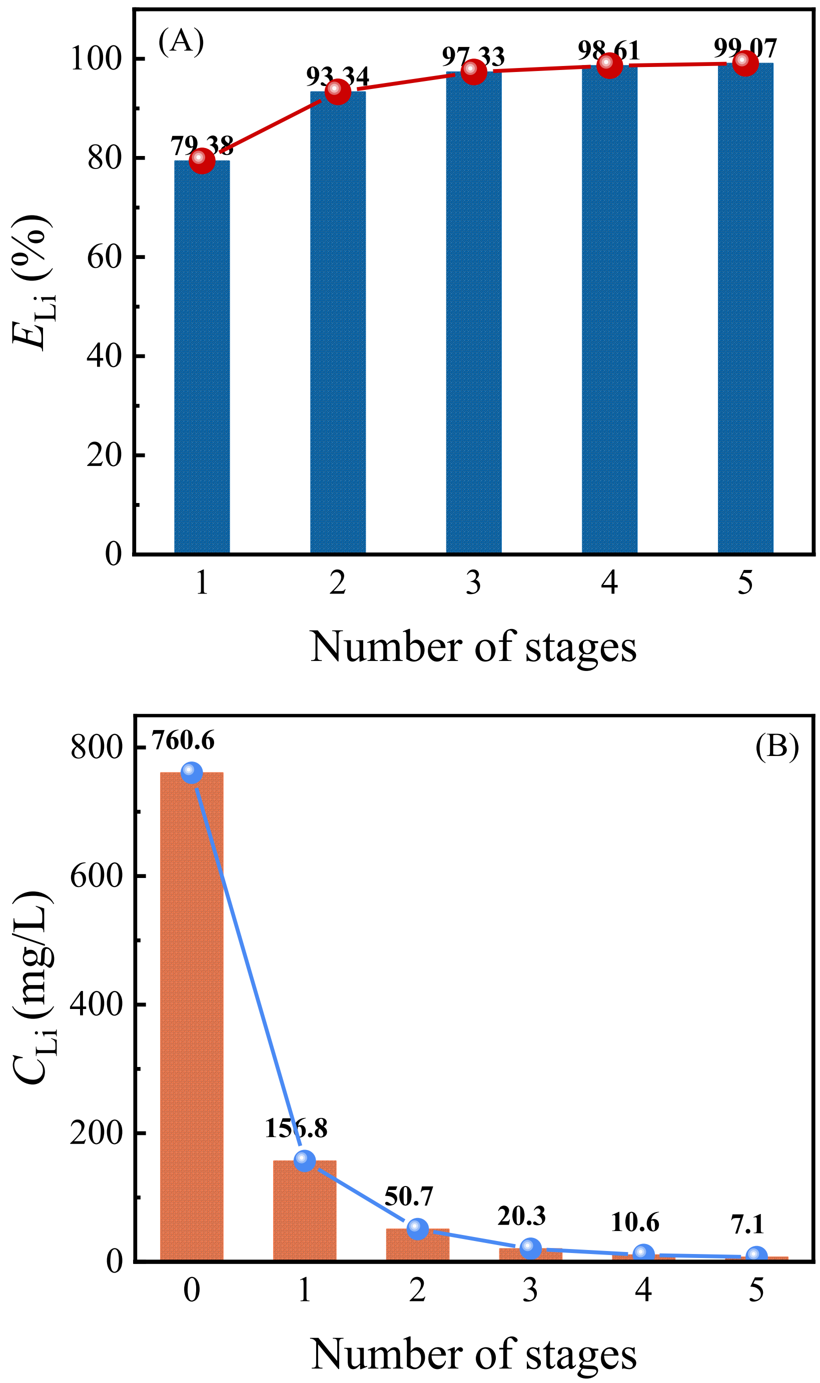

Fig. S2. Extraction rate (A) and concentration of Li^+ (C_{Li+}) in aqueous phase (B) at each stage with [HOEMIM][Tf_2N] + TOP an extractant system (Mg^{2+}/Li^+ =40:1; O/A=3:1; pH = 7 and C_{IL} = 0.1 mol/L) at 298.15 K.

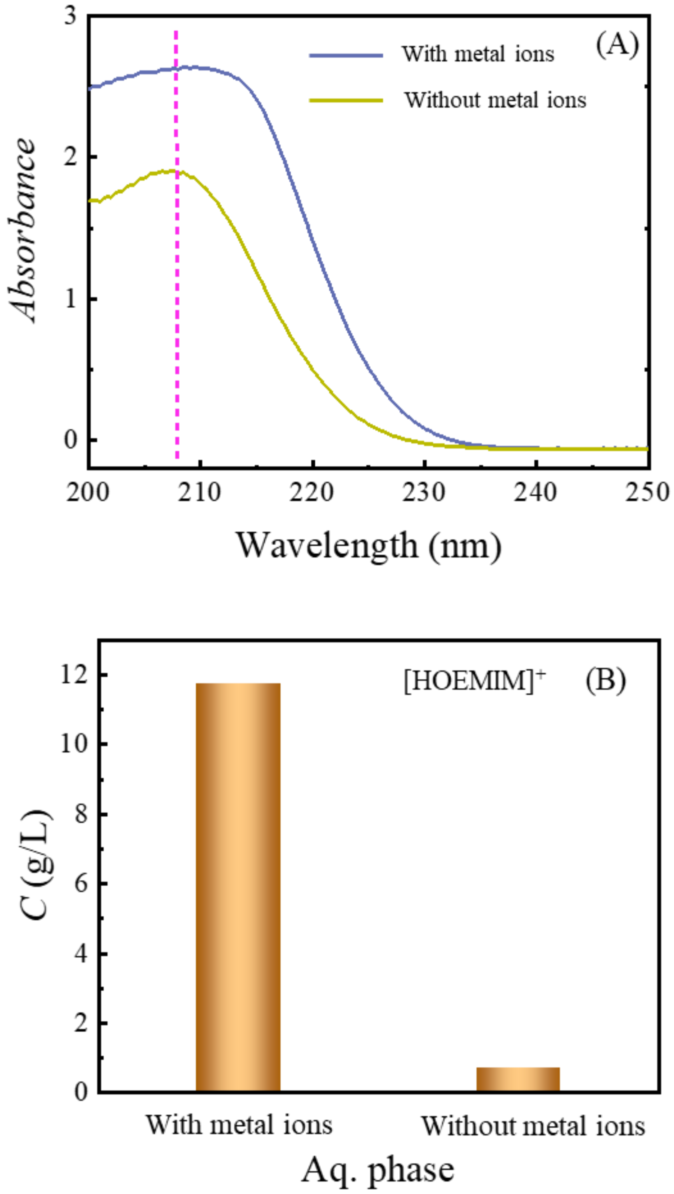

Fig. S3. UV Spectra (A) and $[HOEMIM]^+$ concentrations (B) of the aqueous phase with and without metal ions after extraction with $[HOEMIM][Tf_2N]$ + TOP as extractant system at 298.15 K. The conditions before extraction are in (Li^+: 0.766 g/L, Mg^{2+}: 98.984 g/L, Mg^{2+}/Li^+ =40:1 for the aqueous phase with metal ions before extraction); pH = 7, C_{IL} = 0.09 mol/L, the aqueous phase of 3 mL, and the oil organic phase of 9 mL. $[HOEMIM]^+$ Concentrations are 0.7135 and 11.7477 g/L in the aqueous phase without and with metal ions, respectively, after extraction, and the Li^+ concentration is 0.154 g/L in the aqueous phase with metal ions after extraction.

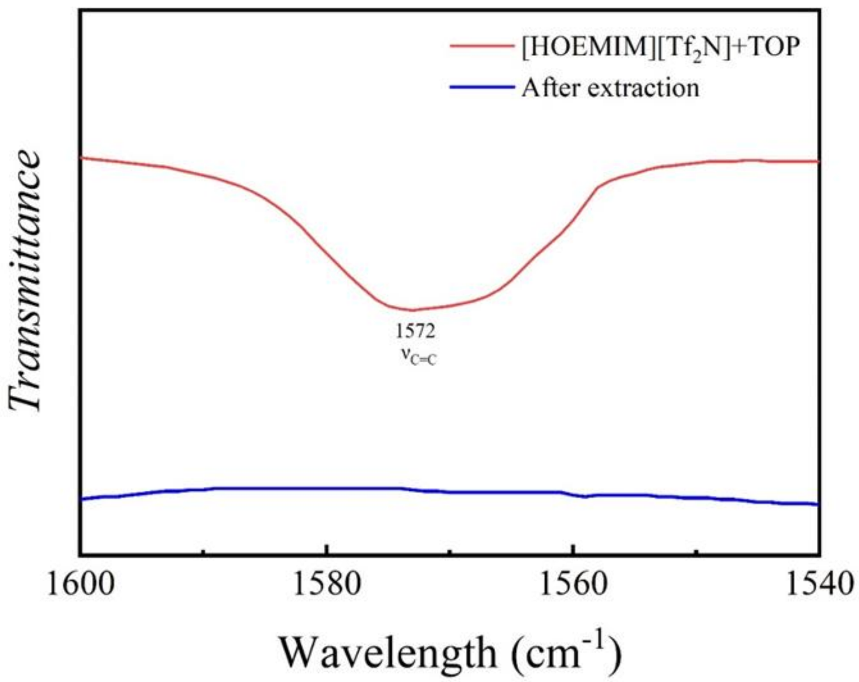

Fig. S4. FTIR spectra of the organic phase before and after extraction at 298.15 K.

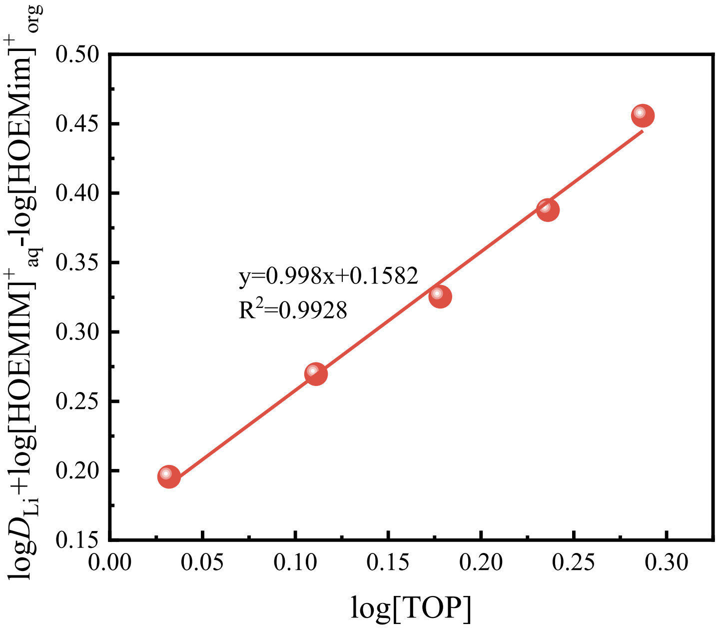

Fig. S5. Plot of $\log D_{Li+}$ + log[HOEMIM]$^+_{aq}$ - log[HOEMIM]$^+_{org}$ as a function of log[TOP] (the Li^+ concentration of 0.1mol/L, O/A = 3:1, C_{IL} = 0.1 mol/L and pH=7) at 298.15 K.